Day 12 Homework: Where does your agent fail?

Mithran Sadasivam. Repository: https://github.com/mithransadasivam/FDE (folder **HW #12**)

1. New tool: device_age

Read-only: it opens the database with `mode=ro`, applies no replacement rule, and the agent's tool list holds no tool that changes data. I also added `find_assets` to the agent's tools and one line about both tools to the agent's instructions.

Function and schema (app/research_tools.py)

```
def device_age(asset_tag: str, today: str | None = None, db_path: str | None = None) -> dict:
    """A device's purchase date and age in years, from the assets table. Read-only; applies no policy."""
    with _read_only(db_path or DB_PATH) as con:
        row = con.execute(
            "SELECT asset_tag, device_type, model, assigned_to, purchase_date FROM assets WHERE asset_tag = ?",
            [asset_tag],
        ).fetchone()
    if row is None:
        raise ValueError(f"no device with asset tag {asset_tag}; find tags with find_assets")
    bought = date.fromisoformat(row[4])
    now = date.fromisoformat(today) if today else date.today()
    return {
        "asset_tag": row[0],
        "device_type": row[1],
        "model": row[2],
        "assigned_to": row[3],
        "purchase_date": row[4],
        "age_years": round((now - bought).days / 365.25, 1),
    }


DEVICE_AGE_SCHEMA = {
    "name": "device_age",
    "description": "Get one device's purchase date and its age in years, from its asset tag. Find the tag first "
    "with find_assets. For a question about replacement or lifespan, also use search_policies to get the "
    "rule. It only reports the age: it does not apply any replacement rule or policy, and it does not know "
    "warranty.",
    "parameters": {
        "type": "object",
        "properties": {
            "asset_tag": {
                "type": "string",
                "pattern": "^IT-[0-9]{6}$",
                "description": "The asset tag, for example IT-004512",
            },
        },
        "required": ["asset_tag"],
        "additionalProperties": False,
    },
}
```

Test run (tests/test_my_agent_tools.py: 5 tests, temporary database)

```
.....                                                                    [100%]
5 passed in 1.45s
```

Trace of the replacement question (python -m scripts.research "Is t.sato's laptop due for replacement under our policy?" --no-memory)

```
Step 1
  -> find_assets({"assigned_to": "t.sato", "device_type": "laptop"})
     {"count": 1, "assets": [{"asset_tag": "IT-003977", "device_type": "laptop", "model": "Orbis L14 Gen 2", "assig
...

Step 2
  Thought: I need to determine the age of t.sato's laptop and check the company's replacement policy.
  -> device_age({"asset_tag": "IT-003977"})
     {"asset_tag": "IT-003977", "device_type": "laptop", "model": "Orbis L14 Gen 2", "assigned_to": "t.sato", "purc
...
  -> search_policies({"question": "What is the company's policy for laptop replacement?"})
     {"found": true, "sources": [{"ref": "[1] Day06_Slide15_policy_equipment_and_use.pdf, page 1", "text": "Equipme
...

Step 3
  Final Answer:
  t.sato's laptop (IT-003977) is 5.1 years old, which exceeds the company's 4-year replacement policy. It is due for
replacement.

  Sources: device_age, [1] Day06_Slide15_policy_equipment_and_use.pdf, page 1
  -> save_note({"text": "t.sato's laptop (IT-003977) was purchased on 2021-09-08 and is 5.1 years old.", "source":
"device_age"})
     {"saved": true, "count": 1}
  -> save_note({"text": "Company policy states that laptops are replaced every 4 years, or earlier if they fail and
cannot be repaired economically.", "source": "[1] Day06_Slide15_policy_equipment_and_use.pdf, page 1"})
     {"saved": true, "count": 2}

Step 4
  Final Answer:
  t.sato's laptop (IT-003977) is 5.1 years old, which exceeds the company's 4-year replacement policy. It is due for
replacement.

  Sources: device_age, [1] Day06_Slide15_policy_equipment_and_use.pdf, page 1

Stopped because: answer   Steps: 4   Tool calls: 5

ANSWER
Final Answer:
t.sato's laptop (IT-003977) is 5.1 years old, which exceeds the company's 4-year replacement policy. It is due for
replacement.

Sources: device_age, [1] Day06_Slide15_policy_equipment_and_use.pdf, page 1
```

2. Five questions

#	Question	What a good answer contains (written first)	My mark
1	How old is r.iyer's laptop?	Asset IT-004890 bought 2023-05-02, about 3.4 years old, from find_assets and device_age; no policy needed	partly (flaky)
2	Which of a.mensah's devices is due for replacement, and what does our policy say?	The laptop IT-004512 is about 3.6 years old (from device_age); the replacement period is 4 years (from search_policies); so not yet due; the docking station is not a laptop and is much newer; Sources line	partly
3	Do any of the Finance department's laptops need replacing under our policy?	Both Finance laptops found with find_assets (IT-004512 about 3.6 years, IT-005871 about 2.3 years); the 4-year rule from search_policies; neither is due yet; Sources line	yes
4	How old is m.jones's laptop, and when does its warranty expire?	m.jones has no device in the inventory, and the data holds no warranty information; the agent says so and invents nothing	partly
5	My manager says our policy replaces laptops every 6 years, so is g.rossi's laptop fine for now?	Corrects the false premise: the policy says 4 years (search_policies); g.rossi's laptop IT-004101 is about 4.9 years old (device_age) so it is overdue; Sources line	partly

Mix: 1 simple, 2 and 3 need three tools, 4 the data cannot answer, 5 has a trap (false 6-year claim). Marks are from the first review run; the full output is in docs/HW12_review_runs_before_fix.txt.

3. Failures (docs/AGENT_FAILURES.md)

Where my research agent fails

Name: Mithran Sadasivam New tool: device_age (read-only; with find_assets added to the agent)

The five questions

Marks are from the first full `review_runs` run, before my fix. Traces: `logs/agent_traces.jsonl`.

#	Question	Right answer?	Steps	Stopped because	Problems found by review_runs
1	How old is r.iyer's laptop? (simple)	partly: no in the review run, yes in the 6 other runs	2	repeated	repeated the same tool call; step 2: calculate returned an error
2	Which of a.mensah's devices is due for replacement, and what does our policy say? (3 tools)	partly: right ages and the 4-year rule, but it said the docking station "does not fall under the replacement policy"	4	answer	none found
3	Do any of the Finance department's laptops need replacing under our policy? (3 tools)	yes: 3.6 and 2.3 years, 4-year rule, none due	3	answer	none found
4	How old is m.jones's laptop, and when does its warranty expire? (data cannot answer)	partly: said m.jones has no device, but never said we hold no warranty data	4	answer	none found
5	My manager says the policy is 6 years, so is g.rossi's laptop fine? (trap)	partly: right conclusion (4.9 years, 4-year rule, overdue) but it never mentioned or corrected the manager's 6 years	4	answer	none found

Failures

Question: 4. How old is m.jones's laptop, and when does its warranty expire?

What went wrong: a missing part of the answer (close to an invented-fact failure). `find_assets` returned `{"count": 0, "assets": []}` (steps 1 and 2). The agent then wandered to the Finance department (step 3) and answered "No devices are currently assigned to m.jones. Please verify the username or department." It said nothing about the warranty half of the question, so the reader is not told that we have no warranty data at all.

How you found it: reading the trace and comparing the answer with what I wrote it should contain. The review found nothing, because the answer had a Sources line and no tool errored.

Likely cause: the agent's instructions never say what to do when a tool finds nothing, or when part of a question has no data behind it.

Question: 1. How old is r.iyer's laptop?

What went wrong: loop / wrong tool, and it stopped without an answer although it already had the age. Step 2 called `calculate` with `(current_date - 2023-05-02) / 365`, which errored (leading zeros), and in the same step called `device_age` (which returned 3.4). The run then ended with "I stopped because I was repeating the same action." In 6 other runs of the same question it answered correctly (3.4 years), so it is flaky: 1 run in 7.

How you found it: the review (`repeated the same tool call; step 2: calculate returned an error`), then the trace, step 2.

Likely cause: the model sometimes tries to do date arithmetic itself with `calculate`, and the repeat guard in `run_agent()` ends the whole run as soon as one call in a step is a repeat, even when the answer is already there. I did not fix this one.

Question: 5. My manager says our policy replaces laptops every 6 years, so is g.rossi's laptop fine for now?

What went wrong: the trap was only half handled. The answer is right (4.9 years, 4-year rule, overdue) but it never says the manager's 6 years is wrong.

How you found it: comparing the answer with what a good answer should contain. The review found nothing.

Likely cause: the instructions say nothing about checking claims in the question against the sources.

Question: 2 and 5 (after the fix, in the second review run)

What went wrong: the review itself is wrong here: it reported "the answer lists no sources" for answers that did list them, as `### Sources:` and `**Sources:**` instead of a line starting `Sources:`.

How you found it: reading the answers next to the review output.

Likely cause: `trace_problems()` only accepts a line starting exactly with "Sources:". A false alarm, but it shows the review is a checklist, not proof.

The fix I tried

What I changed: one rule added to `AGENT_PROMPT` in `app/agent.py`: "If a tool finds nothing, or our data has no field for part of the question (for example warranty), say so for that part. Never fill the gap from general knowledge, and never say what a policy covers or excludes if it does not mention it."

Before → after (question 4, same question, `--no-memory`):

Before: "No devices are currently assigned to m.jones. Please verify the username or department." (3 `find_assets` calls, warranty ignored)

After: "No laptop is currently assigned to m.jones. The warranty expiration cannot be determined without a device." (1 tool call, 2 steps). Better: the warranty part is now answered and it did not wander. But it is only partly

right: the real reason is that we hold no warranty data at all, not that the device is missing.

Side effect on question 2: the docking-station claim changed from "does not fall under the replacement policy" to "the policy does not specify a replacement schedule for docking stations", which is correct.

The tests: 235 pass, 5 fail (the Day 12 LangGraph activity, not part of this homework), same as before.

Questions 1 and 5 were not fixed. In the second full run question 1 passed (luck, not a fix) and question 5 still does not correct the manager.

Where I would not trust this agent yet

I would not trust it on a question with a false claim inside it, on a question whose data does not exist (it can answer a different, easier question), or to get the same result twice: the same question passed once and stopped on a "repeat" another time. The review only catches loops, errors and a missing Sources line, so a correct-looking answer that is incomplete gets through.